

Zihan Zhou (周孜涵)

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Education

Ph.D. in Computer Science

- **Johns Hopkins University** 08/2025–Present
- **Advisor:** [Murat Kocaoglu](#)
- **Selected coursework:** Database, Computational Genomics

Ph.D. Studies in Electrical and Computer Engineering

- **Purdue University** 08/2021–08/2025
- Transferred to Johns Hopkins University with advisor [Murat Kocaoglu](#)
- **Advisors:** [Somali Chaterji](#) 08/2021–05/2023
[Murat Kocaoglu](#) 08/2023–08/2025
- **Selected coursework:** Intro to AI, Statistical Machine Learning, Introduction to Neural Networks, Probabilistic Causal Inference, Graph Theory, Bayesian Data Analysis, Real Analysis and Measure Theory

M.S. in Electrical Engineering

- **Northwestern University** 09/2019–03/2021
- **Advisor:** [Thrasyvoulos Pappas](#)
- **Selected coursework:** Advanced Computer Vision, Intro to Robotics, Machine Learning, Deep Learning, Stochastic Process, Numerical Methods in Nonlinear System Control Optimization

B.S. in Space Science and Technology

- **Nanjing University** 09/2015–06/2019
- **Research advisors:** [Liyong Zhou](#), [Hui Zhang](#)
- **Selected coursework:** Linear Algebra, Calculus, Control Theory, Celestial Mechanics, Theoretical Mechanics, Numerical Methods, Mathematical Methods for Physics

Research Interests

Causal machine learning, causal discovery, Bayesian causal inference, root-cause analysis, reliable and interpretable AI, multi-agent AI systems

Publications

Peer-Reviewed Publications

- Yang Xu, Jiefu Zhang, Haixiang Sun, **Zihan Zhou**, Tianyu Cao, and Vaneet Aggarwal. “Towards Reliable LLM Evaluation: Correcting the Winner’s Curse in Adaptive Benchmarking.” Advances in Neural Information Processing Systems (NeurIPS), 2026.
- Haixiang Sun, Pengchao Tian, **Zihan Zhou**, and Andrew Liu. “Hierarchical Causal Structure Discovery from Layered Topological Orderings.” International Journal of Data Science and Analytics (IJDSA), 2026.
- Kenneth Lee, **Zihan Zhou**, and Murat Kocaoglu. “Root Cause Analysis of Failures in Microservices via Bayesian Root Cause Discovery.” International Conference on Machine Learning (ICML), 2026. **Spotlight**.
- **Zihan Zhou** and Murat Kocaoglu. “Towards Completeness in Causal Discovery from Soft Interventions with Known Targets.” International Conference on Machine Learning (ICML), 2026.
- **Zihan Zhou**, Muhammad Qasim Elahi, and Murat Kocaoglu. “Characterization and Learning of Causal Graphs from Hard Interventions.” Advances in Neural Information Processing Systems (NeurIPS), 2025.
- **Zihan Zhou**, Muhammad Qasim Elahi, and Murat Kocaoglu. “Sample Efficient Bayesian Learning of Causal Graphs from Interventions.” Advances in Neural Information Processing Systems (NeurIPS), 2024.
- Karthick Shankar, Ashraf Mahgoub, **Zihan Zhou**, Utkarsh Priyam, and Somali Chaterji. “Asgard: Are NoSQL Databases Suitable for Ephemeral Data in Serverless Workloads?” Frontiers in High Performance Computing, 2023.
- **Zihan Zhou**, Zijia Du, and Somali Chaterji. “Kratos: Context-Aware Cell Type Classification and Interpretation Using Joint Dimensionality Reduction and Clustering.” In Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2022.
- Atul Sharma, Pranjal Jain, Ashraf Mahgoub, **Zihan Zhou**, Kanak Mahadik, and Somali Chaterji. “Lerna: Transformer Architectures for Configuring Error Correction Tools for Short- and Long-Read Genome Sequencing.” BMC Bioinformatics, 2022.

Submitted / Under Review / Preprints

- **Zihan Zhou** and Murat Kocaoglu. “Causal Discovery from Diverse Data.” Under review.
- Haixiang Sun, Yang Xu, Jiefu Zhang, Xudong Wu, **Zihan Zhou**, Jun He, Jiayu Chen. “Core-Halo Decomposition: Decentralizing Large-Scale Fixed-Point Problems.” Preprint, arXiv:2605.08681.
- Haixiang Sun, Pengchao Tian, **Zihan Zhou**, Jielei Zhang, Peiyi Li, and Andrew Liu. “Efficient Causal Structure Learning via Modular Subgraph Integration.” Preprint, arXiv:2601.21014.

Selected Research Experience

Johns Hopkins University, Causal Machine Learning Lab

Advisor: [Murat Kocaoglu](#)

- **Causal Discovery from Diverse Data** Dec 2025–Present
 - Developed a causal discovery framework for heterogeneous datasets whose differences need not admit an interventional interpretation.
 - Derived cross-domain invariance constraints and an inv-Markov equivalence characterization using F-node augmented graphs.
 - Proposed a constraint-based learning algorithm combining within-domain conditional independence tests with cross-domain invariance tests.
 - Submitted to ICLR 2027.
- **Towards Complete Learning of Causal Graphs from Soft Interventions** May 2025–Jan 2026
 - Studied identifiability and complete learning of causal graphs from soft-interventional data.
 - Developed graphical and algorithmic tools for reasoning about mechanism changes induced by soft interventions.
 - Accepted to ICML 2026.
- **Bayesian Root-Cause Discovery for Microservice Failures** Jan 2025–Oct 2025
 - Co-developed a Bayesian root-cause discovery method for identifying failure sources from limited abnormal samples.
 - Outperformed state-of-the-art baselines on synthetic and real-world microservice datasets.
 - Started at Purdue.
 - Accepted to ICML 2026 as a Spotlight paper.

Independent and Collaborative Research Projects

- **Core-Halo Decomposition for Decentralized Fixed-Point Problems** Jan 2026–May 2026
 - Co-developed a decentralized decomposition framework that separates disjoint update ownership from overlapping read-only evaluation context.
 - Proved exact fixed-point recovery under a locality condition and characterized the structural bias of strict local decomposition.
 - Evaluated the framework on MiniGrid navigation, PageRank, smart-grid energy management, and multi-intersection traffic control.
 - Preprint, arXiv:2605.08681.
- **Towards Reliable LLM Evaluation** Dec 2025–May 2026
 - Co-developed SIREN, a selection-aware repeated-split protocol for reporting LLM benchmark results after adaptive prompt and program search, targeting the fresh-data performance of the full tune-then-deploy procedure rather than the observed winner’s score.
 - SIREN freezes the post-search shortlist, separates splitwise selection from held-out evaluation, and quantifies uncertainty with an item-level Gaussian multiplier bootstrap, with guarantees for simultaneous inference over a finite grid of tuning budgets.
 - Experiments on MMLU-Pro with random-search and DSPy tuners across eleven open-weight LLMs show that winner-based reporting can be optimistic and can change deployment conclusions.
 - Accepted to NeurIPS 2026.
- **Hierarchical Causal Structure Discovery** May 2025–Sep 2025

- Co-developed a scalable causal discovery algorithm that integrates local orderings over Markov blanket-induced subgraphs.
- Improved both scalability and accuracy over state-of-the-art causal discovery baselines in empirical evaluations.
- Published in IJDSA 2026.
- **Efficient Causal Structure Learning via Modular Subgraph Integration** Jan 2025–May 2025
 - Introduced a modular framework that decomposes global causal structure learning into local subgraph learning problems based on Markov blankets.
 - Co-designed a weighted voting mechanism for global graph integration that penalizes low-support edges.
 - Established finite-sample error bounds and proved asymptotic consistency under mild conditions.
 - Achieved improved accuracy and efficiency on synthetic and real datasets across a range of base learners.

Purdue University, Causal Machine Learning Lab

- **Causal Discovery from Hard Interventional Data** Jun 2024–May 2025
Advisor: [Murat Kocaoglu](#)
 - Proposed a graphical representation for hard-interventional Markov equivalence classes.
 - Developed a sound causal discovery algorithm for learning from multiple hard-interventional datasets.
 - Accepted to NeurIPS 2025.
- **Sample-Efficient Bayesian Causal Discovery** Sep 2023–May 2024
Advisor: [Murat Kocaoglu](#)
 - Developed a sample-efficient Bayesian algorithm for learning causal graphs from interventional data.
 - Provided asymptotic and non-asymptotic theoretical analyses of the proposed method.
 - Accepted to NeurIPS 2024.

Purdue University, Innovatory for Cells and Neural Machines

- **Transformer-Based Configuration of K-mer Error Correction** Jun 2021–Sep 2021
Advisor: [Somali Chaterji](#)
 - Co-developed a transformer-based approach for automatically configuring k-mer-based error correction tools for genome sequencing datasets.
 - Leveraged language-model representations to optimize k-mer size selection across sequencing settings.
- **Cell Type Classification and Interpretation for scRNA-seq Data** Oct 2021–May 2023
Advisor: [Somali Chaterji](#)
 - Developed an interpretable machine learning framework for joint dimensionality reduction, cell type classification, and marker gene extraction.
 - Published in KDD 2022.

- **Adaptive Cold-Start Mitigation for Serverless Cloud Computing** Jun 2022–May 2023
Advisor: [Somali Chaterji](#)

- Co-developed a serverless computing framework that balances cold-start reduction, execution time, energy efficiency, and resource usage.
- Co-designed a hybrid policy that leverages DAG-based workload predictability for optimized cloud and edge performance.

Northwestern University, Department of Electrical and Computer Engineering

- **Perceived Dynamic Range on Dead Leaves Models** Jun 2020–Mar 2021
Advisor: [Thrasyvoulos Pappas](#)

- Generated synthetic images using dead leaves models to study perceived dynamic range without semantic context.
- Designed controlled image-generation experiments for perceptual image-quality analysis.

Nanjing University, School of Astronomy and Space Science

- **CNN-Based Recognition of Astronomical Transient Sources** Jun 2018–May 2019
Advisor: [Hui Zhang](#)

- Built an image-stamp dataset from sky survey image subtraction outputs using SExtractor.
- Trained a convolutional neural network to recognize astronomical transient sources.

- **Orbital Effects of a Hypothetical Outer Perturber on Asteroids** Jun 2017–Mar 2018
Advisor: [Liyong Zhou](#)

- Simulated long-term asteroid dynamics under different initial orbital parameters.
- Analyzed the results using numerical methods based on the Lagrange perturbation equations.

Teaching

- (Purdue) TA of ECE 57000 Intro to AI, Spring 2025
- (Purdue) TA of ECE 62900 Intro to Neural Network, Fall 2024
- (Purdue) TA of ECE 50024 Machine Learning, Spring 2024
- (Purdue) TA of ECE 57000 Intro to AI, Fall 2023
- (Purdue) TA of ABE 59100 Machine Learning for CV and IoT, Spring 2023

Academic Service

- Reviewer: UAI 2024–2026; ICLR 2025, 2027; AISTATS 2025–2026; NeurIPS 2025–2026; ICML 2026

Awards

- ICML 2026 Silver Reviewer Award (2026)
- NeurIPS 2024 Scholar Award (2024)
- KDD 2022 Student Volunteer (2022)
- People's Scholarship (2018)
- Scholarships for Incoming Freshmen (2015)

Certificates

- [AWS Cloud Technical Essentials](#), Amazon Web Services
- [Natural Language Processing Specialization](#), DeepLearning.AI
- [Deep Learning Specialization](#), DeepLearning.AI
- [Image and Video Processing: From Mars to Hollywood with a Stop at the Hospital](#), Duke University
- [Divide and Conquer, Sorting and Searching, and Randomized Algorithms](#), Stanford University
- [Data-driven Astronomy](#), The University of Sydney
- [Machine Learning](#), Stanford University

Skills

- Languages: Mandarin, English, Japanese
- Programming Languages: Python, Matlab, C++, Julia, R
- Machine Learning Libraries: PyTorch, TensorFlow, pyAgrum, OpenCV
- Others: AWS, MySQL